

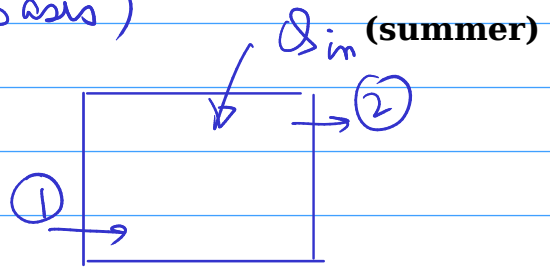
Case A

energy and mass balance

(per kg dry air basis)

Room

Process 1-2

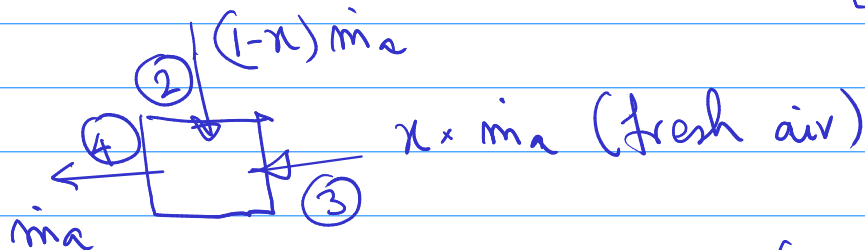


$$Q_{in} = \dot{m}_a (h_2 - h_1)$$

dry air
 $\left[\frac{\text{kg}}{\text{s}} \right]$

water vap addition = $\dot{m}_a (w_2 - w_1)$
in $\left[\frac{\text{g of moisture}}{\text{kg of dry air}} \right]$

Mixer



Adiabatic mixing (i.e., no heat addition)

$$(1-x)\dot{m}_a \cdot h_2 + x\dot{m}_a h_3 = \dot{m}_a \cdot h_4$$

(Note: neglect KE & PE)

cooler

Process 4-5

t_A is "Apparatus dew point"

(cooling coil)

$$\text{efficiency } \eta = \frac{t_4 - t_5}{t_4 - t_A}$$

moisture removed =

$$\dot{m}_a h_4 = \dot{m}_a h_5 + Q_{out \text{ cooling}}$$

$$\dot{m}_a (w_4 - w_5)$$

heater

Process 5-6

$$\dot{m}_a h_5 + Q_{in} = \dot{m}_a h_6$$

↑ heating (Power = $I \times V$)

fan

Process 6-1

$$\dot{m}_a h_6 + W_{in} = \dot{m}_a h_1$$

↑ work in (power = $I \times V$)

case-2

This is a case where the air in the room is getting cooled as it is winter and there is net sensible heat loss from the room.

Room

Process 1-2

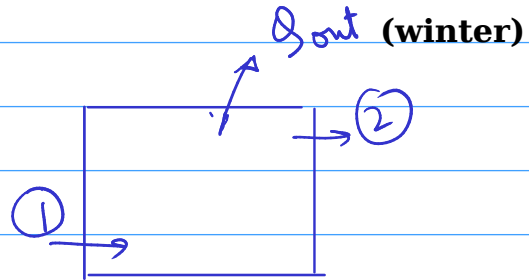
$$Q_{out} = \dot{m}_a (h_1 - h_2)$$

dry air
[$\frac{\text{kg}}{\text{s}}$]

water vap

$$\text{addition} = \dot{m}_a (w_2 - w_1)$$

in [$\frac{\text{g of moisture}}{\text{kg of dry air}}$]



③: specified air

